REVISION 2021 • SEMESTER 6 • TEXTILE TECHNOLOGY

Statistical Quality Control in Textile Testing

A full-screen, syllabus-style digital handbook for Course Code 6061B. It is written as a practical textbook with module explanations, Malayalam support notes, formulas, checklists, worked cases, lab/report guidance, revision questions, model question paper and full answer key.

COURSE CODE
6061B

COURSE TITLE
Statistical Quality Control in Textile Testing

DEPARTMENT
Textile Technology

TYPE
Theory

SEMESTER
6

CATEGORY
Program Elective

USE
Study + notes PDF

FORMAT
Big handbook

6061B

Screen-filling handbook

Designed for desktop, mobile, classroom display and automatic PDF notes generation.

How to use this handbook

Read the overview first, then study each module with the diagram, formulas, practical tasks and model answers. For exam preparation, revise the question bank and answer key.

Course map	**objectives + outcomes**	**Module lessons**	**4 detailed units**
Formula bank	**calculations**	**Practice bank**	**questions + tasks**
Model paper	**official style**	**Answer key**	**full points**

Course objectives and syllabus map

Objectives

- ✓ Use statistical tools to interpret textile testing data
- ✓ Apply control charts, sampling and process capability
- ✓ Connect test results to textile production and quality decisions

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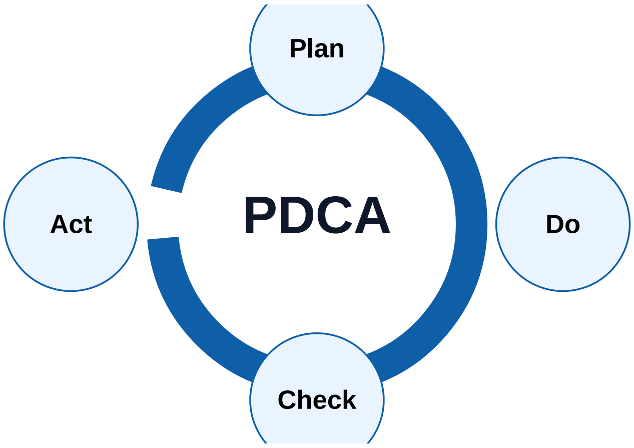
Visual process map



Module	Main outcome	Industrial focus	Expected evidence
M1	Statistics for textile testing	Need for statistics in fibre, yarn and fabric testing; Data types, sample, population, frequency distribution and histogram	Short notes, calculations, diagram, checklist, viva answers
M2	Sampling and testing reliability	Random sampling, lot sampling and sample size selection; Testing errors: instrument, operator, atmospheric and material variation	Short notes, calculations, diagram, checklist, viva answers
M3	Control charts and process control	Chance causes and assignable causes; X-bar and R chart, p-chart, np-chart, c-chart and u-chart	Short notes, calculations, diagram, checklist, viva answers
M4	Acceptance sampling and capability	AQL, lot acceptance and defect classification; OC curve idea and producer/consumer risk	Short notes, calculations, diagram, checklist, viva answers

Module-wise textbook

Each module is written in clear engineering language. Do not mug up headings only; understand the process flow, defects, records and decision points.

Module 1

Statistics for textile testing

Core explanation

- ✓ Need for statistics in fibre, yarn and fabric testing
- ✓ Data types, sample, population, frequency distribution and histogram
- ✓ Mean, median, mode, range, variance, standard deviation and coefficient of variation
- ✓ Normal distribution and interpretation of variation

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Exam writing method

1. Start with definition or purpose.
2. Draw block diagram/process flow wherever possible.
3. List parameters, machines/tools, defects and precautions.
4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 2

Sampling and testing reliability

Core explanation

- ✓ Random sampling, lot sampling and sample size selection
- ✓ Testing errors: instrument, operator, atmospheric and material variation
- ✓ Confidence, tolerance and repeatability
- ✓ Outlier identification and reporting discipline

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Common mistakes

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- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 3

Control charts and process control

Core explanation

- ✓ Chance causes and assignable causes
- ✓ X-bar and R chart, p-chart, np-chart, c-chart and u-chart
- ✓ Upper and lower control limits and process stability
- ✓ Applications in count, strength, CSP, fabric defects and shade variation

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 4

Acceptance sampling and capability

Core explanation

- ✓ AQL, lot acceptance and defect classification
- ✓ OC curve idea and producer/consumer risk
- ✓ Process capability Cp and Cpk
- ✓ Using SQC for mill improvement, customer complaint reduction and audit evidence

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

□□□□□□: Practical answer □□□□□□□□□□ definition □□□□□□ □□□□. Process → parameter → defect → correction → record □□□□ flow □□□□□□□□□.

Exam writing method

1. Start with definition or purpose.
2. Draw block diagram/process flow wherever possible.
3. List parameters, machines/tools, defects and precautions.
4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Formula bank and quick calculations

Formula 1

Mean = Sum of observations / n

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 2

Range = Maximum - Minimum

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 3

CV % = Standard Deviation / Mean x 100

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 4

UCL/LCL = Centre Line ± 3 sigma

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 5

Cp = (USL - LSL) / 6 sigma

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Practice tasks, viva and revision

Practical / assignment tasks

- ✓ Calculate mean, SD and CV for yarn count readings
- ✓ Prepare X-bar and R chart for tensile strength
- ✓ Interpret AQL result for fabric defects
- ✓ Prepare a test report with conclusion

Important keywords

Mean **SD** **CV** **Control chart** **AQL** **Cp** **Cpk** **Sampling**

Short questions

Q1.1	Explain: Need for statistics in fibre, yarn and fabric testing	Answer with definition, process, application and one example.
Q1.2	Explain: Data types, sample, population, frequency distribution and histogram	Answer with definition, process, application and one example.
Q1.3	Explain: Mean, median, mode, range, variance, standard deviation and coefficient of variation	Answer with definition, process, application and one example.
Q2.1	Explain: Random sampling, lot sampling and sample size selection	Answer with definition, process, application and one example.
Q2.2	Explain: Testing errors: instrument, operator, atmospheric and material variation	Answer with definition, process, application and one example.
Q2.3	Explain: Confidence, tolerance and repeatability	Answer with definition, process, application and one example.
Q3.1	Explain: Chance causes and assignable causes	Answer with definition, process, application and one example.

Q3.2	Explain: X-bar and R chart, p-chart, np-chart, c-chart and u-chart	Answer with definition, process, application and one example.
Q3.3	Explain: Upper and lower control limits and process stability	Answer with definition, process, application and one example.
Q4.1	Explain: AQL, lot acceptance and defect classification	Answer with definition, process, application and one example.
Q4.2	Explain: OC curve idea and producer/consumer risk	Answer with definition, process, application and one example.
Q4.3	Explain: Process capability Cp and Cpk	Answer with definition, process, application and one example.

Case study

A textile unit receives customer complaint related to Mean. Prepare a corrective action report using observation, data collection, root cause, corrective action, preventive action and verification.

Expected answer structure: Problem statement → data/table → cause analysis → action plan → responsible person → completion date → verification evidence.

Model question paper

Use this as a sample paper. Questions are original and arranged in diploma-style sections.

PART A - Short Answer Questions (answer all)
1. Define Statistical Quality Control in Textile Testing.
2. List four important keywords: Mean, SD, CV, Control chart.
3. State two industrial applications of this subject.
4. Mention two common defects/problems and one preventive action.
5. Write any one important formula or calculation used in this subject.

PART B - Paragraph Questions
6. Explain the workflow of Statistics for textile testing.
7. Compare two important concepts from Sampling and testing reliability.
8. Explain the practical importance of Control charts and process control in a textile unit.
9. Write the steps for documentation/reporting in this subject.

PART C - Essay / Application Questions
10. Prepare a complete process plan for Statistical Quality Control in Textile Testing in a textile industry situation.
11. Explain how quality, cost, safety and customer requirement are controlled in this subject.
12. Solve/interpret a simple industrial case using the formulas and checks given in this handbook.

Answer key and scoring points

Part A answer points

1. Give accurate definition and scope of Statistical Quality Control in Textile Testing.
2. Use keywords: Mean, SD, CV, Control chart, AQL, Cp, Cpk, Sampling.
3. Applications must be linked to textile industry, customer requirement and quality.
4. Defects/problems should include cause and prevention.
5. Formula answers must show unit and interpretation.

Part B/C - Module 1

Statistics for textile testing

- Need for statistics in fibre, yarn and fabric testing
- Data types, sample, population, frequency distribution and histogram
- Mean, median, mode, range, variance, standard deviation and coefficient of variation
- Normal distribution and interpretation of variation

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 2

Sampling and testing reliability

- Random sampling, lot sampling and sample size selection
- Testing errors: instrument, operator, atmospheric and material variation
- Confidence, tolerance and repeatability
- Outlier identification and reporting discipline

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 3

Control charts and process control

- Chance causes and assignable causes
- X-bar and R chart, p-chart, np-chart, c-chart and u-chart
- Upper and lower control limits and process stability
- Applications in count, strength, CSP, fabric defects and shade variation

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 4

Acceptance sampling and capability

- AQL, lot acceptance and defect classification
- OC curve idea and producer/consumer risk
- Process capability Cp and Cpk
- Using SQC for mill improvement, customer complaint reduction and audit evidence

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Final quick revision

Before exam

- Revise all formulas and keywords.
- Practise one diagram/process flow from every module.
- Prepare two industrial examples for every long answer.

Before lab/viva

- Know instrument/machine purpose.
- Know safety precautions and standard record format.
- Explain result, defect and correction logically.